

Clustering Formula 1 Lap Data Using K-means, DBSCAN, and Hierarchical Clustering

Winnie Trinh

Dept. of Computer Science
California Polytechnic State University
San Luis Obispo, CA, USA
witrinh@calpoly.edu

Nathan Madlansacay

Dept. of Statistics
California Polytechnic State University
San Luis Obispo, CA, USA
nmadlans@calpoly.edu

Abstract—This project analyzes Formula 1 data to determine whether lap-level driving patterns can be utilized to discover natural groupings among teams and laps. We began by building our dataset from the 2023 Formula 1 qualifying session data from the OpenF1 API. We also gathered session information, driver metadata, lap data, stint data, weather data, and car samples. For consistency, rainy sessions, pit-out laps, and incomplete laps were excluded. Only competitive laps within 107% of the session’s best lap time were kept. We used DuckDB to join lap records with samples and engineer lap-level features like throttle behavior, braking percentage, gear usage, RPM statistics, speed ranges, DRS usage, and number of gear changes. Team and driver labels were attached to the final cleaned dataset.

After finalizing the dataset, we applied three clustering methods: K-means clustering, DBSCAN, and hierarchical agglomerative clustering. The clusters were evaluated using silhouette score, Rand Index, radius/intercluster distance ratio, and outlier counts. PCA was used to visualize the results. Of the three clustering methods, DBSCAN performed best overall with the highest silhouette score of 0.479 and the highest Rand Index of 0.848. 42 laps were identified as outliers, suggesting that some laps had unusual patterns. The results show that F1 contains meaningful structure. However, the clusters and team labels do not match exactly, suggesting that lap behavior may be affected by driver style, track conditions, and car performance.

Index Terms—Formula 1, data mining, clustering, K-means clustering, DBSCAN, hierarchical clustering, PCA visualization, outlier detection

I. INTRODUCTION

Formula 1 is a data-heavy sport, and each lap contains information about driver behavior, car performance, and track conditions. Since teams and drivers make different strategic and mechanical choices, data provides an opportunity for discovery. We wanted to investigate whether Formula 1 lap data contains enough structure to group laps in meaningful ways; we were especially interested in whether unsupervised clustering methods could separate laps by team or reveal other driving behavior patterns.

First, we collected and prepared a Formula 1 dataset using the 2023 qualifying session data from the OpenF1 API. We wanted to create our own dataset so that we had the desired variables and format for further calculations. This was done by combining session, driver, lap, stint, weather, and car information. The data was filtered to focus on competitive dry laps by removing incomplete or less reliable laps. Lap-

level features such as throttle behavior, braking percentage, RPM statistics, speed range, DRS usage, gear usage, and gear changes were constructed. To prevent any variables with large numeric ranges from dominating the clustering algorithms, we standardized all of the features.

After finalizing the dataset, we then applied three clustering methods: K-means clustering, DBSCAN, and hierarchical agglomerative clustering. K-means clustering partitions the data into a fixed number of clusters, DBSCAN groups points based on density and can identify outliers, and hierarchical clustering builds a nested structure of clusters. We were able to compare the discovered clusters against the actual teams using Rand Index because the dataset included true team labels. We also evaluated cluster quality using silhouette score and radius/intercluster distance ratio. Lastly, we used PCA to project the data into two dimensions for visualization.

Our goal was to determine which clustering method performed best as well as to understand what the clustering results suggest about Formula 1. If clusters closely matched team labels, this would suggest that team-level car characteristics strongly influence lap patterns. If the clusters did not match the team perfectly, this would suggest that other factors, like driver style, track conditions, or individual lap behavior, also play an important role.

II. DATASET

We created a Formula 1 dataset using data from the OpenF1 API [1]. The dataset focused on the 2023 Formula 1 qualifying sessions because qualifying laps are competitive laps since drivers are trying to achieve the fastest possible time. This makes qualifying data useful for studying lap-level performance patterns. The raw data collected included session information, driver information, lap data, stint data, weather data, and car samples.

The first step in creating the finalized dataset was collecting the raw data from the OpenF1 API and saving it into parquet files. For each qualifying session, we retrieved the session metadata, weather data, driver list, lap records, stint records, and car data for each driver. Rainy sessions were excluded because wet conditions can significantly affect how one performs on the track, making them harder to compare fairly with dry laps. We loaded the parquet files into DuckDB tables so the different

data sources could be joined and queried [4].

The next step was to remove unimportant or non-competitive laps. Pit-out laps were removed as they do not represent normal full-speed laps. Laps with missing times or unrealistic durations were also removed. We kept laps that were within 107% of the best lap time from their session to focus on competitive laps. This helped remove any laps that could potentially distort the clustering results.

Then, we joined the lap records with the car samples. Each sample was assigned to the lap it occurred during by matching the session key, driver number, and timestamp range between the start and end of the lap. We formed features that summarized driver and car behavior through these samples. Such features included average throttle, throttle standard deviation, percent of time at full throttle, percent of time at zero throttle, percent of time braking, average gear, number of unique gears used, average RPM, RPM standard deviation, maximum RPM, maximum speed, average speed, minimum speed, speed standard deviation, percentage of time in different speed ranges, percentage of time with DRS active, and number of gear changes. Team and driver labels were then attached to each lap by joining the feature table with the driver table. The team label was used as a label for later evaluation on whether the clusters discovered by the algorithms matched the actual Formula 1 teams. The final dataset was saved as `lap_features_labeled.csv`.

Before performing the clustering algorithms, we removed non-feature columns such as team name, driver number, session key, lap number, and lap starting time from the clustering input. Data cleaning and feature preparation were also supported using pandas and NumPy [2], [3]. The other features were standardized using StandardScaler so that features measured on different scales would contribute more fairly to the distance calculations [6]. The final scaled dataset was saved as `f1_laps_scaled.csv`, with the true team label stored separately for later evaluations.

III. ANALYTICAL QUESTIONS

The goal of this project was to determine whether Formula 1 data contains meaningful patterns that can be discovered through clustering. Each lap contains information about driver behavior and car performance, so we wanted to see whether laps would naturally group based on their features. The main question we studied was whether Formula 1 laps could be clustered into groups that correspond to actual teams. Although team labels were included in the final labeled dataset, they were not used as input features during clustering. Instead, the clustering algorithms used numeric features like throttle behavior, braking percentage, speed, RPM, gear usage, DRS usage, and gear changes. The team labels were used afterwards to evaluate how closely the discovered clusters matched the true team values.

Another question we were trying to answer was which clustering method worked best for this dataset. We compared K-means clustering, DBSCAN, and hierarchical agglomerative clustering. K-means was used to test whether the data could

be divided into ten clusters, matching the ten Formula 1 teams. DBSCAN was used to determine whether the data contained density-based clusters and possible outlier laps. Hierarchical clustering was used to observe how laps grouped together based on similarity.

The project focused on the following questions:

- 1) Can Formula 1 data naturally cluster laps into meaningful groups?
- 2) Do the discovered clusters correspond to actual Formula 1 teams?
- 3) Which clustering method performs best on this dataset?
- 4) Can DBSCAN identify unusual laps or outliers in the data?
- 5) What do the clustering results suggest about the relationship between the teams and lap behavior?

IV. METHODS

Three clustering methods were applied to the Formula 1 dataset: K-means clustering, DBSCAN, and hierarchical agglomerative clustering. These methods were chosen because each one groups data in a different way. K-means requires a fixed number of clusters, DBSCAN forms clusters based on density and can detect outliers, and hierarchical clustering builds a tree-like structure of clusters. Through comparison, we can study which algorithm best groups laps based on behavior and how closely those groups match the actual Formula 1 teams.

A. K-means Clustering

K-means clustering is an unsupervised clustering method that separates data points into a fixed number of clusters [8]. First, the algorithm selects initial centroids, then assigns each data point to the closest centroid based on distance. After all points are assigned, the centroids are then recomputed as the average of the points in the cluster. This process is repeated and stops when the centroids stop changing significantly. In this project, we used $k = 10$ since the dataset contained laps from 10 Formula 1 teams. This allowed us to test whether the data could be divided into 10 groups that corresponded to the actual teams. However, this method has its limitations. It assumes the number of clusters is known ahead of time, and it works best when clusters are compact and relatively evenly shaped.

B. DBSCAN

DBSCAN is a density-based clustering algorithm that groups together points located in dense regions of the feature space [7]. A fixed number of clusters is not required. Its two main parameters are epsilon and minimum points such that epsilon defines the radius around a point and minimum points defines how many neighbors are needed for a point to be considered part of a dense region.

In this project, we used $\epsilon = 2.0$ and minimum points = 10. This method was able to identify clusters without forcing the data into exactly 10 groups as well as label unusual points as noise. Points were labeled as (-1) if they were outliers in our project.

C. Hierarchical Agglomerative Clustering

Hierarchical agglomerative clustering is an algorithm that builds groups step by step. It starts with each data point as its own cluster, then repeatedly merges the closest clusters until all points are combined into one hierarchy. This can be represented as a dendrogram, which shows how clusters are merged at different distance levels.

In this project, we used average linkage hierarchical clustering, where the distance between two clusters is measured using the average distance between points in each cluster. This method also does not require a fixed number of clusters. After the hierarchy is built, the dendrogram is cut at a threshold of 5.0 to produce final cluster labels.

D. PCA Visualization

The original dataset contained many numeric features, so it could not be directly visualized in two dimensions. To create readable plots, we used Principal Component Analysis (PCA), which reduces high-dimensional data into a smaller number of dimensions while preserving as much variation in the data as possible.

In this project, PCA was used for visualization. All three clustering algorithms were run on the full standardized feature set rather than the two-dimensional PCA data. After clustering, PCA was used to project the laps into two dimensions so that we could plot the clustering results and compare them with the true team labels [9].

V. EVALUATION METRICS

We used several evaluation metrics to compare the clustering methods, including silhouette score, Rand Index, radius/intercluster distance ratio, and outlier count [6]. Clustering is an unsupervised learning method, so the algorithms were not trained using the true team labels. Since the dataset still included the actual team labels, we were able to use them afterward to evaluate how closely the discovered clusters matched the real Formula 1 teams.

A. Silhouette Score

The silhouette score measures how well each point fits within its assigned cluster compared to other clusters. A greater silhouette score means that points are closer to points in their own cluster and further from points in other clusters. This allows us to evaluate the separation and compactness of the clusters without relying on the true team labels.

B. Rand Index

The Rand Index is used to compare the predicted cluster labels to the actual team labels. This helped measure how similar the clustering results were to the true team groupings. A greater Rand Index means that the clustering result matched the team labels more closely.

C. Radius/Intercluster Distance Ratio

The radius/intercluster distance ratio was used to measure how compact and separated the clusters were. The radius measures how spread out the points are within a cluster, and the intercluster distance measures how far apart different clusters are. The smaller the ratio, the better, as it signifies that the clusters are relatively compact and well-separated.

D. Outlier Count

The outlier count for DBSCAN was useful as it showed how many laps had unusual patterns that did not fit clearly into any main cluster. Unlike K-means and hierarchical clustering, DBSCAN can label points as noise, and in our project, those points were labeled as (-1) and treated like outliers.

VI. RESULTS

After applying K-means clustering, DBSCAN, and hierarchical agglomerative clustering to the Formula 1 dataset, we compared the results using silhouette score, Rand Index, radius/intercluster distance ratio, and outlier count. The final dataset contained 1,502 lap records. Table I summarizes the results for each clustering method.

TABLE I
COMPARISON OF CLUSTERING RESULTS

Method	Silhouette	Rand Index	Radius Ratio	Outliers
DBSCAN	0.479	0.848	1.117	42
Hierarchical	0.432	0.775	0.722	N/A
K-means	0.384	0.789	2.016	N/A

Overall, DBSCAN performed the best based on the main evaluation metrics. It had the highest silhouette score of 0.479, meaning its clusters were the most separated compared to the other methods. It also had the highest Rand Index value of 0.848, signifying that its cluster assignments matched the true team labels more closely than K-means or hierarchical clustering. DBSCAN also identified 42 laps as outliers, meaning these laps had patterns that did not fit clearly into one of the dense clusters.

K-means produced exactly 10 clusters to match the number of Formula 1 teams in the dataset. Its silhouette score was 0.384, and its Rand Index was 0.789, which were both lower than DBSCAN. This suggests that forcing the data into 10 clusters did not capture the structure of the data as well as a density-based approach.

Hierarchical clustering produced a silhouette score of 0.432 and a Rand Index of 0.775. Its radius/intercluster distance ratio was 0.722, making it the lowest amongst the three. This suggests that hierarchical clustering created relatively compact and separated clusters. However, its Rand Index was lower than DBSCAN and K-means, meaning its clusters did not match the actual team labels as closely.

The PCA visualizations helped interpret the results and were plotted using Matplotlib [5]. Figure 1 shows how K-means divided the laps into 10 groups after projecting the high-dimensional data into two dimensions. Figure 2 shows that the

actual teams overlapped in the PCA space instead of forming separate groups. This supports the numeric results because the clusters did not perfectly match the true Formula 1 teams.

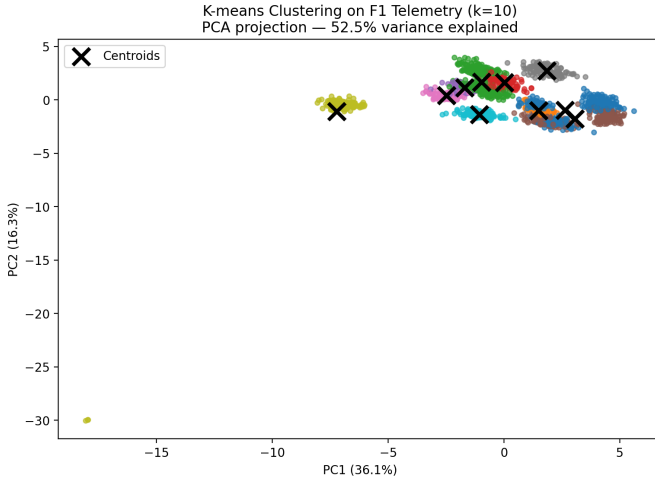


Fig. 1. K-means clustering results projected into two dimensions using PCA.

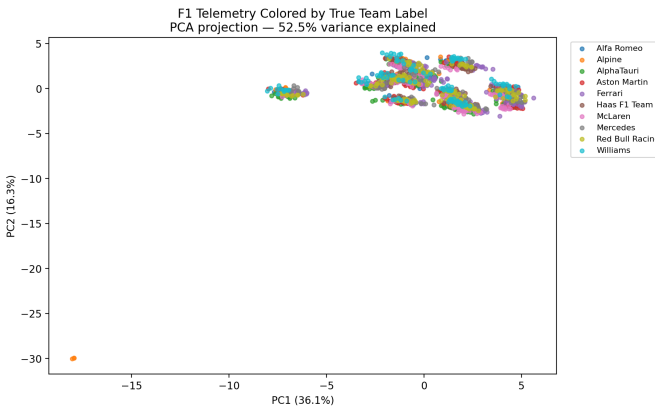


Fig. 2. True Formula 1 team labels projected into two dimensions using PCA.

VII. DISCUSSION

The results suggest that the Formula 1 data contains meaningful structure, but the structure does not perfectly correspond to the actual team labels. If the features separated cleanly by team, we would expect the clustering methods to produce groups that closely match the 10 Formula 1 teams. However, the results show that the clusters only partially matched the team labels. This suggests that team identity is important, but it is not the only factor affecting lap patterns.

DBSCAN performed the best overall since it had the highest silhouette score and the highest Rand Index. This means that its clusters were more separated and more similar to the true team labels than the clusters produced by K-means or hierarchical clustering. DBSCAN may have performed better because the Formula 1 data may not have formed perfectly round or evenly sized clusters. K-means works best when the data can be divided into compact groups, but patterns vary

in shape and density. Since DBSCAN groups points based on density, it is better at finding natural patterns in this type of data.

The K-means results show that forcing the data into 10 clusters did not fully capture the dataset's structure. The data did not separate into 10 clear groups, suggesting that laps from different teams can still look similar based on speed, throttle, braking, RPM, and gear usage. Also, laps from the same team may not always look identical since drivers, car setup, and lap-specific conditions can affect behavior.

DBSCAN labeled 42 laps as outliers, meaning those laps did not fit clearly into any dense cluster. These laps may represent unusual patterns such as abnormal braking, throttle usage, speed behavior, or other lap conditions. Even though the dataset was cleaned to remove incomplete or noncompetitive laps, some unusual laps still remained.

The PCA visualizations support these findings. The true team label plot showed the teams overlapped in the two-dimensional PCA space instead of forming separate groups, meaning team identity alone does not explain all the variation in behavior. How the laps were grouped was likely influenced by other factors like driver style, track characteristics, car setup, and session conditions.

VIII. CONCLUSION

In this project, we built and analyzed a Formula 1 dataset to study whether lap-level driving patterns could be grouped using clustering methods. After collecting and preparing the data, we engineered features related to speed, throttle, braking, RPM, gear usage, DRS usage, and gear changes. The numeric features were then standardized, and three methods were applied: K-means clustering, DBSCAN, and hierarchical agglomerative clustering.

Overall, DBSCAN performed the best among the three methods, having the highest silhouette score and Rand Index. This suggests that its clusters were more separated and similar to the true team labels than the clusters produced by the other two methods. DBSCAN also identified 42 outlier laps, showing that some laps had unusual patterns that did not fit clearly into the main clusters.

The results suggest that the Formula 1 data does contain meaningful structure, but the clusters do not perfectly correspond to the 10 team labels. This means that while team identity and car performance influence patterns, other factors such as driver style, car setup, track conditions, and lap-specific behavior also play an important role. In future work, this project could be expanded to include race sessions, multiple seasons, tire information, sector times, and track-specific features to further understand the factors that shape Formula 1 lap behavior.

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